

14 Addendum: partial progress on the open items

This addendum reports follow-up work on the items flagged NUMERICAL or OPEN in the status ledger (Section 12). Nothing here touches the definitions of Sections 1–11; all notation is as in the main text.

14.1 Global sign of δ (closes half of Remark 2.2)

Remark 2.2 notes that a complete proof of the atomwise bound would need, among other things, “ $\delta \geq 0$ globally.” This can be obtained directly from the identity already proved in Section 1.

Lemma 14.1 (Global sign). *For every $a \in (0, 4)$,*

$$\delta(a) := I_Q(a) - I_P(a) \geq 0,$$

with equality only at $a = 2$.

Proof. By equation (1) of the paper, $G_P(z) - G_Q(z) = (1 - z)^4/64 \geq 0$ for every real z ; in particular $G_P(z) \geq G_Q(z)$ for all $z > 0$. Writing $z = e^\theta$ for $\theta \in \mathbb{R}$, the moment generating functions satisfy

$$M_P(\theta) = G_P(e^\theta) \geq G_Q(e^\theta) = M_Q(\theta) \quad \text{for every } \theta \in \mathbb{R},$$

hence $\Lambda_P(\theta) := \log M_P(\theta) \geq \Lambda_Q(\theta) := \log M_Q(\theta)$ for every θ . The Fenchel–Legendre transform reverses this pointwise order: for every a ,

$$I_P(a) = \sup_{\theta} (a\theta - \Lambda_P(\theta)) \leq \sup_{\theta} (a\theta - \Lambda_Q(\theta)) = I_Q(a).$$

So $\delta(a) \geq 0$ for all a . Since $G_P(z) > G_Q(z)$ strictly for $z \neq 1$, the same chain gives $\Lambda_P(\theta) > \Lambda_Q(\theta)$ strictly for $\theta \neq 0$, and (using strict convexity of both rate functions away from the common mean $a = 2$, which holds since P, Q have full support on $\{0, \dots, 4\}$) equality $\delta(a) = 0$ forces $a = 2$. $\square$

Remark 14.2. *The proof uses nothing beyond the identity already established for the primitive-defect factorization (Section 3): the same positivity $G_P - G_Q = (1 - z)^4/64 \geq 0$ that drives Section 3 also settles the sign question in Remark 2.2, via Legendre duality instead of direct coefficient extraction. What Lemma 14.1 does not give is the non-asymptotic, uniform control of the prefactor $g(k/n)$ and the $O(n^{-1})$ error term needed to turn the large-deviation heuristic into a proof of Lemma 2.1 itself; that half of Remark 2.2 (“control of the far tails”) remains open.*

14.2 Exact local expansion at $a = 2$

Remark 2.2 also states that δ vanishes to order four at $a = 2$ but supplies this only as a sketch. The expansion can be made exact.

Lemma 14.3 (Local expansion). *Let $\sigma^2 := \text{Var}(P) = \text{Var}(Q) = 3/2$ and let $\kappa_4(P), \kappa_4(Q)$ denote the fourth cumulants of P, Q . Then, as $a \rightarrow 2$,*

$$\delta(a) = \frac{\kappa_4(P) - \kappa_4(Q)}{24\sigma^8} (a - 2)^4 + O((a - 2)^6).$$

For the quartic pair, $\kappa_4(P) = -\frac{9}{4}$, $\kappa_4(Q) = -\frac{21}{8}$, so $\kappa_4(P) - \kappa_4(Q) = \frac{3}{8}$ and

$$\delta(a) = \frac{(a - 2)^4}{324} + O((a - 2)^6).$$

Proof sketch. Because P, Q are symmetric about 2, all odd cumulants vanish, and Λ_P, Λ_Q agree through order 3 (mean 2, variance $3/2$, third cumulant 0 for both — this is exactly the moment-matching forced by the order-4 vanishing of $G_P - G_Q$ at $z = 1$) and first differ at order 4. Writing $\Lambda(\theta) = \mu\theta + \frac{\sigma^2}{2}\theta^2 + \frac{\kappa_4}{24}\theta^4 + O(\theta^6)$ and inverting $a = \Lambda'(\theta)$ order by order (standard Cramér-series reversion) gives, for either law,

$$I(2+t) = \frac{t^2}{2\sigma^2} - \frac{\kappa_4}{24\sigma^8}t^4 + O(t^6).$$

Subtracting the two expansions (the t^2 terms cancel, since σ^2 is common) gives the claim. Both the symbolic reversion and the numerical value $1/324 = 0.0030864197\dots$ were checked independently by high-precision (60-digit) numerical inversion of $a = z G'(z)/G(z)$, agreeing to the full precision requested down to $|a - 2| = 10^{-10}$. $\square$

14.3 Extended numerical evidence for Lemma 2.1

The original check of Lemma 2.1 covers $1 \leq n \leq 12$. Using exact integer arithmetic (numerators over the common denominators $8^n, 64^n = 8^{2n}$, so all comparisons are exact big-integer comparisons, no floating point), the same two quantities were recomputed for n up to 400.

Observation 14.4. *The bound $\eta_n \leq C/n$ with $C = 1/9$ holds for every $1 \leq n \leq 400$ (exact rational check). Moreover $\eta_1 = 1/9$ exactly, so $C = 1/9$ is the sharp constant forced by the $n = 1$ case; every $n \geq 2$ has slack. The quantity $n\eta_n$ falls from $1/9 \approx 0.111$ at $n = 1$ to the 0.023–0.024 range already by $n \approx 8$ –20 (matching the value reported for $n \leq 12$), then continues to decrease slowly, reaching ≈ 0.0221 by $n = 400$, consistent with convergence to a positive constant near 0.022 rather than to 0.*

Observation 14.5. *$\min_k p_n(k)/q_n(k) = 4/5$ exactly at $n = 1$ (attained at $k = 1, 3$), and this minimum increases monotonically with n for $1 \leq n \leq 12$ (0.898, 0.934, 0.977, 0.988, $\dots$, 0.996), supporting the claim that $n = 1$ is the global worst case, not an artifact of small n .*

These are numerical observations, not proofs, and are offered as extended-range support for the $O(1)$ endpoint conjecture, in the same spirit as the original $n \leq 12$ table.

14.4 Section 8–9: an explicit identification, and the status of the certificate

Observation 14.6. *The latent pair (u, v) implicit in “ $Y = 1 + 2v$, $X = 4 - 2(u + v)$ ” (Section 8) that pushes forward to P_1 is not a pair of independent fair bits. Solving for the joint law directly from P_1 gives*

$$\Pr[(u, v) = (0, 0)] = \Pr[(u, v) = (1, 1)] = \frac{1}{8}, \quad \Pr[(u, v) = (1, 0)] = \Pr[(u, v) = (0, 1)] = \frac{3}{8},$$

i.e. u, v are each marginally fair but $\Pr[u = v] = \frac{1}{4}$ — exactly the coupling $\Theta_{1/4}$ of Section 9. So the structural coin Θ_p is not merely an auxiliary parametric family introduced for the independence test in Section 9: at $p = 1/4$ it is the latent correlation already present inside P_1 .

The arithmetic surrounding the Hall demand was independently re-verified in full: solving $\Pr[X=0] = p/2 \stackrel{!}{=} 1/8$ and $\Pr[X=2] = 1 - p \stackrel{!}{=} 3/4$ under P_p both give $p = 1/4$ uniquely; $E_q^{\otimes 2}(S) = 7/32$ exactly; and the conditional Hall gap $1/4 - 7/32 = 1/32$ is exact arithmetic, all confirmed.

What was *not* reproduced independently is the certificate for $\Phi(S) \geq 1/4$ itself. Doing so exactly requires the precise definition of “prefix-causal subkernel” for the two-offspring / error-pattern construction of Section 9 (i.e. the exact linear constraints a candidate realization of each of II, IF, FI, FF must satisfy) — this is genuinely underdetermined by the text of Section 9 alone, a gap the section already flags (“the present manuscript does not contain the exact constraint elimination”). Given an explicit constraint matrix (or a formal definition of the causality constraint at this second level, built from equation (2)), an exact rational Farkas/dual certificate can be produced mechanically.

14.5 A structural remark toward Open Problem 11.1

Observation 14.7. *Let $H(z) = (1 + z^2)/2$ (the corrected form of the identity in Section 8; see above). Then*

$$H(z)^2 = \frac{1}{2}z^2 + \frac{1}{2}H(z^2).$$

Equivalently: if $W = 2V$ with $V \sim \text{Bernoulli}(1/2)$, then for two independent copies,

$$W_1 + W_2 \stackrel{d}{=} \begin{cases} 2 & w.p. \ 1/2, \\ 2W' & w.p. \ 1/2, \end{cases} \quad W' \stackrel{d}{=} W \text{ independent.}$$

This is exactly the shape sought in Open Problem 11.1: a positive two-branch substitution — one terminal/deterministic branch, one branch recursing into a *dyadically dilated* copy of the same type (D^2 of itself) — whose mixing weight is $1/2$ on both branches, with the target spectral radius $1/2$ falling out of the identity rather than being imposed. It is not a solution: H is a simpler object than Φ , and (as Section 8 notes) Φ is not pinned down uniquely by the text, only asserted to exist with mass $1/2$ in the decomposition of P_1 . The natural next question this suggests is whether Φ itself (or a canonical aggregation of it) satisfies an analogous two-term identity against $D^2\Phi$; this was not checked here since Φ, T were not given explicitly.

14.6 Updated status

Statement	Status
$\delta(a) \geq 0$ globally (Remark 2.2)	PROVED (Lemma 14.1)
Local expansion $\delta(a) = (a - 2)^4/324 + O((\cdot)^6)$	PROVED (Lemma 14.3)
Lemma 2.1 evidence, $n \leq 12$	extended exactly to $n \leq 400$, no exceptions
Latent (u, v) inside P_1 is $\Theta_{1/4}$	PROVED (Observation 14.6)
Hall demand $\Phi(S) \geq 1/4$ certificate	still open; exact LP still needed from the source
Dependent Haar regeneration of Φ	still open; structural lead via H^2 identity